Number 1 — Fractions · Order of operations · Rounding

No calculator. Give answers as fractions or mixed numbers in their lowest terms unless the question says otherwise. Marks are shown in brackets. ★ = harder question (grades 7–9).

A · FRACTIONS

1 Simplify [3]

a $\frac{18}{48}$ **b** $\frac{150}{48}$ **c** $\frac{2.4}{48}$

a Divide top and bottom by the HCF, 6: $\frac{18}{48} = \frac{3}{8}$.

b $\frac{150}{48} = \frac{25}{8} = 3\frac{1}{8}$ (HCF 6).

c Multiply top and bottom by 10 first: $\frac{2.4}{48} = \frac{24}{480} = \frac{1}{20}$.

2 Work out [4]

a $\frac{4}{9} \times 1\frac{1}{8}$ **b** $3\frac{1}{3} \div \frac{5}{6}$

a Convert to an improper fraction, then cancel before multiplying:
$\frac{4}{9} \times \frac{9}{8} = \frac{4 \times 9}{9 \times 8} = \frac{4}{8} = \frac{1}{2}$.

b To divide, multiply by the reciprocal: $\frac{10}{3} \div \frac{5}{6} = \frac{10}{3} \times \frac{6}{5} = \frac{60}{15} = 4$.

3 Work out $\frac{5}{6} + 2\frac{3}{4} - 1\frac{2}{3}$ [3]

Improper fractions, then a common denominator (LCM of 6, 4, 3 is 12):
$\frac{5}{6} + \frac{11}{4} - \frac{5}{3} = \frac{10}{12} + \frac{33}{12} - \frac{20}{12} = \frac{23}{12} = 1\frac{11}{12}$.

4 In a school, $\frac{2}{5}$ of the students study Spanish, $\frac{1}{4}$ study French, $\frac{1}{10}$ study German and the rest study no language. [5]

a What fraction of the students study no language?

b 90 students study no language. How many students are in the school?

a Common denominator 20: $\frac{8}{20} + \frac{5}{20} + \frac{2}{20} = \frac{15}{20}$, so the rest is $1 - \frac{15}{20} = \frac{5}{20} = \frac{1}{4}$.

b $\frac{1}{4}$ of the school is 90 students, so the school has $90 \times 4 = 360$ students.

5 A jug holds $3\frac{3}{4}$ litres of juice. A cup holds $\frac{3}{8}$ litre. How many full cups can be poured from the jug? [3]

> "How many $\frac{3}{8}$ in $3\frac{3}{4}$" is a division: $\frac{15}{4} \div \frac{3}{8} = \frac{15}{4} \times \frac{8}{3} = \frac{120}{12} = 10$ cups.

B · ORDER OF OPERATIONS

6 Work out [4]

a $5 + 3 \times (2+3)^2$ **b** $\frac{2}{3} + \frac{1}{3} \div \frac{4}{9}$

> **a** Brackets, then indices, then multiply, then add:
> $5 + 3 \times 5^2 = 5 + 3 \times 25 = 5 + 75 = 80$.
> **b** Division before addition: $\frac{1}{3} \div \frac{4}{9} = \frac{1}{3} \times \frac{9}{4} = \frac{3}{4}$, so
> $\frac{2}{3} + \frac{3}{4} = \frac{8}{12} + \frac{9}{12} = \frac{17}{12} = 1\frac{5}{12}$.

7 Insert one pair of brackets to make this calculation correct: $2 + 3 \times 4 - 1 = 19$ [2]

> Without brackets the value is $2 + 12 - 1 = 13$. Try forcing the addition first:
> $(2+3) \times 4 - 1 = 5 \times 4 - 1 = 19$ ✓.

8 Evaluate $\left(1\frac{1}{2} - \frac{2}{3}\right)^2 \div \left(\frac{5}{6} + \frac{1}{4}\right)$ [3]

> First bracket: $\frac{9}{6} - \frac{4}{6} = \frac{5}{6}$, squared: $\frac{25}{36}$. Second bracket: $\frac{10}{12} + \frac{3}{12} = \frac{13}{12}$.
> $\frac{25}{36} \div \frac{13}{12} = \frac{25}{36} \times \frac{12}{13} = \frac{25}{39}$.

C · SIGNIFICANT FIGURES AND DECIMAL PLACES

9 The height of Mount Everest is $8\,848.86$ m. Write this height correct to [2]

a 1 decimal place **b** 2 significant figures

> **a** The 2nd d.p. is 6 (≥ 5), so round up: $8\,848.9$ m.
> **b** The first two s.f. are 8 and 8; the 3rd s.f. is 4 (< 5), so leave them: $8\,800$ m. (The zeros keep the size of the number.)

10 $\pi = 3.14159265\ldots$ Write π correct to [2]

a 4 significant figures **b** 4 decimal places

> **a** 4 s.f.: look at the 5th s.f., 5 → round up: 3.142.
> **b** 4 d.p.: look at the 5th d.p., 9 → round up: 3.1416.

11 The mass of a grain of pollen is $0.000\,004\,09$ g. [3]

a Write this mass correct to 1 significant figure.

b Write it correct to 6 decimal places.

c Explain why your answer to **b** is less useful than your answer to **a**.

> **a** The first s.f. is the 4; the next digit is 0, so $0.000\,004$ g.
>
> **b** 6 d.p.: $0.000\,004\,09 \to 0.000\,004$ (the 7th d.p. is 0). Same digits, but…
>
> **c** …to 6 d.p. the answer has only one non-zero digit left, so it says nothing about the 9; s.f. follow the size of the number, d.p. do not.

D · MIXED PROBLEMS

12 A bag of rice weighs $2\frac{2}{5}$ kg. [5]

a A café uses $\frac{3}{10}$ kg of rice each day. For how many full days does the bag last?

b A different café uses $\frac{1}{3}$ of a new bag in the first week and then $\frac{3}{4}$ of what is left in the second week. What mass of rice is left after two weeks?

> **a** $\frac{12}{5} \div \frac{3}{10} = \frac{12}{5} \times \frac{10}{3} = \frac{120}{15} = 8$ days.
>
> **b** After week 1 the fraction left is $1 - \frac{1}{3} = \frac{2}{3}$. Week 2 uses $\frac{3}{4}$ of that, so $\frac{1}{4}$ of it remains: $\frac{2}{3} \times \frac{1}{4} = \frac{1}{6}$ of the bag. Mass: $\frac{1}{6} \times \frac{12}{5} = \frac{2}{5}$ kg.

13 Kai says: "-3^2 and $(-3)^2$ are the same because both contain -3 and a square." Is Kai right? Work out both values and explain. [3]

> Indices come before the minus sign (which here means $-1\times$): $-3^2 = -(3 \times 3) = -9$. With brackets the square applies to -3: $(-3)^2 = (-3) \times (-3) = 9$. Kai is wrong — the bracket changes which operation is done first.

Total: 42 marks

Check: fractions in lowest terms · mixed numbers where the answer is bigger than 1 · units in word problems.